



Harvest
HUB

SMART AGRICULTURE USING IoT

Intelligent Farming for a
Sustainable Future

learn more



Challenges Faced by Farmers

- 🌱 Water Wastage
 - Excessive irrigation leads to unnecessary water consumption.
- 🌱 No Real-Time Monitoring
 - Farmers cannot continuously monitor soil moisture and weather conditions.
- 🌱 Low Crop Yield
 - Improper irrigation and delayed decisions affect productivity.
- 🌱 High Labor Requirement
 - Manual farming requires more time and effort.



Need for Smart Agriculture

An IoT-based smart farming system helps farmers monitor crops in real time, save water, and improve crop productivity.



Proposed Solution



Soil Moisture Monitoring

Measures soil moisture levels in real time to determine when irrigation is needed.



Automatic Irrigation

The water pump turns ON/OFF automatically based on soil moisture readings.



Weather Monitoring

Monitors temperature and humidity using IoT sensors for better farming decisions.



Mobile App Alerts

Farmers receive real-time updates and notifications through a mobile application.





Key Features

Project Overview

Our IoT-based Smart Agriculture System continuously monitors field conditions using smart sensors. It automates irrigation, reduces water wastage, and helps farmers make better decisions through real-time monitoring.

Goal

To improve crop productivity, conserve water, reduce manual effort, and promote sustainable farming using IoT technology.



Real-Time Monitoring

- Soil Moisture Sensor
- Temperature Monitoring
- Humidity Monitoring
- Rain Detection



Smart Automation

- Automatic Irrigation
- Mobile Notifications
- Water Level Monitoring
- Cloud Data Storage

Technology Stack

Hardware

- ESP32 / NodeMCU
- Soil Moisture Sensor
- DHT11 Sensor



Software

- Arduino IDE
- Firebase
- Blynk IoT
- HTML, CSS & JavaScript



Additional Components

- Rain Sensor
- Relay Module
- Water Pump



Cloud & Communication

- Wi-Fi Module
- Cloud Database
- Mobile Application
- Real-Time Notifications



System Architecture

Sensor Data Collection

- Soil Moisture Sensor
- DHT11 Temperature & Humidity Sensor
- Rain Sensor

Data Processing

The ESP32/NodeMCU receives sensor data and sends it to the cloud using Wi-Fi.

Smart Irrigation & Monitoring

- Mobile app displays live data.
- Water pump turns ON/OFF automatically.
- Farmers receive instant notifications.

Benefits of Smart Agriculture Using IoT

Our IoT-based smart agriculture system improves farming efficiency by automating irrigation, monitoring crop conditions in real time, and helping farmers make better decisions.

Water Conservation

- Reduces water wastage through automatic irrigation.
- Uses water only when soil moisture is low.

Increased Crop Yield

- Maintains optimal soil conditions.
- Helps improve crop growth and productivity.

Saves Time & Labor

Fluctuating market prices make it hard for farmers to plan and invest. One bad season or market crash can cause major losses.

Sustainable Farming

- Conserves natural resources.
- Promotes eco-friendly and efficient agriculture.

Future Scope



AI-Based Disease Detection

Use Artificial Intelligence to identify crop diseases at an early stage through leaf images, helping farmers take quick action.



Drone Monitoring

Use drones to monitor crop health, detect pests, and survey large agricultural fields efficiently.



Smart Analytics & Prediction

Analyze sensor and weather data to predict crop yield, recommend irrigation schedules, and improve farming decisions.

Expected Outcomes

Our Smart Agriculture Using IoT system improves farming by automating irrigation, monitoring crop conditions in real time, and helping farmers increase productivity while conserving natural resources.



Water Conservation

Reduces water wastage
by 30–40%
Uses water only when
required

Higher Crop Yield

Improves plant growth
Increases overall
productivity

Reduced Farming Cost

Less manual labor
Efficient use of water
and electricity

Sustainable Agriculture

Eco-friendly farming
Better resource
management
Supports long-term
agricultural growth



Harvest Hub – Smart Agriculture
Solution

Thank you for your time and attention.

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Any Questions?

